Third Semester MGU-UGP (Honours) Regular Examination October 2025
Discipline Specific Core Course (DSC) - MG3DSCECT201
Discipline Specific Elective (DSE) - MG3DSEECC200

Programming in C

(2024 Admission onwards)
Maximum Marks: 50
Answer Key and Scheme

Part A – Multiple Choice Questions (1 mark each)

Q.No	Question Topic	Correct Answer
1	Increment/decrement in for loop	**(d) both b and c**
2	Function prototype	**(b) It declares the function to the compiler before use**
3	Partial structure initialization	**(b) Remaining members are set to 0 (zero-initialized)**
4	Purpose of problem analysis	**(b) To identify the inputs, processes, and outputs**
5	Index of first element in array	**(b) 0**
6	Syntax to end a structure	**(a) with a semicolon (;)**
7	Control structure for menu-driven program	**(c) switch case**
8	Default return type of function	**(b) int**
9	Use of pointers, arrays, and strings	**(b) Accessed and manipulated more efficiently**
10	Initial value of struct members	**(d) random garbage values**

Total: 10 marks

Part B – Short Answer Questions (5 marks each)

11. Arithmetic operators in C

Listing of three operators: 2 marks
Explaining the function of each: 1 mark × 3 = 3 marks

- + → addition
- − → subtraction
- * → multiplication
- / → division
- % → modulus (remainder)

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12. Local and global variables with same name

The local variable takes precedence (shadowing) – 1 mark
Illustration: 1 mark

```
#include <stdio.h>
int x = 10; // global
int main() {
    int x = 20; // local
    printf("%d", x); // prints 20
    return 0;
}
```

Example program: 3 marks

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13. Declaring pointer variables

A pointer holds the address of a variable of the specified type – 2 marks
Declaration – 3 marks

```
int *p;      // pointer to int
char *ch;    // pointer to char
float *fp;   // pointer to float
```

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14. Role of header files

Role of header files: They allow reuse of predefined library functions – 1 mark

- `<stdio.h>`: standard input/output functions (`printf()`, `scanf()`) – 2 marks
- `<math.h>`: mathematical functions (`sqrt()`, `pow()`, `sin()`) – 2 marks

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15. Pointer initialization

Pointers must be initialized before dereferencing – 1 mark
Initialization example: 4 marks

```
int a = 5;
int *p = &a;  // p initialized with address of a
```

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16. Program using `continue` to print odd numbers 1–20

Program with proper usage of ‘continue’, correct logic but with errors – 4 marks
Logically correct program without errors – 5 marks

```
#include <stdio.h>
int main() {
    for (int i = 1; i <= 20; i++) {
        if (i % 2 == 0)
            continue;
        printf("%d ", i);
    }
```

```
    return 0;
}
```

Part C – Essay Questions (10 marks each)

17. C Tokens and Types

Definition: C tokens are the smallest individual elements of a C program – 2 marks
Listing of any four – 2 marks
Listing with examples – 4 marks
Listing with examples and explanation – 8 marks

Types:

1. **Keywords** – e.g., `int`, `return`, `if`
2. **Identifiers** – variable, function, array names
3. **Constants** – e.g., `10`, `'A'`, `3.14`
4. **Strings** – e.g., `"Hello"`
5. **Operators** – `+`, `-`, `*`, `/`, `==`
6. **Special symbols** – `{}`, `()`, `;`, `,`, etc.

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18. Program to copy one string to another

Program with proper usage of 'strcpy()', correct logic but with errors – 8 marks
Logically correct program without errors – 10 marks

```
#include <stdio.h>
#include <string.h>
int main() {
    char src[100], dest[100];
    printf("Enter a string: ");
    gets(src);
    strcpy(dest, src);
    printf("Copied string: %s", dest);
    return 0;
}
```

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19. Program demonstrating function pointer

Function pointers allow calling functions dynamically – 2 marks
Program – 6
Step by step explanation – 2 marks

```
#include <stdio.h>
```

```
void display(int n) {
    printf("Number: %d\n", n);
}

int main() {
    void (*fp)(int);  // function pointer declaration
    fp = display;     // assigning function address
    fp(10);           // calling function through pointer
    return 0;
}
```

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20. Program to check palindrome string

Program with correct logic but with errors – 8 marks
Logically correct program without errors – 10 marks
(Input – 3, Process – 5, Output – 2)

```
#include <stdio.h>
#include <string.h>

int main() {
    char str[100];
    printf("Enter a string: ");
    gets(str);
    int len = strlen(str), flag = 1;
    for (int i = 0; i < len / 2; i++) {
        if (str[i] != str[len - i - 1]) {
            flag = 0;
            break;
        }
    }
    if (flag)
        printf("Palindrome");
    else
        printf("Not a palindrome");
    return 0;
}
```